Abdullah AL Maruf

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I am a Python instructor and research mentor working remotely in India, specializing in Machine Learning, Deep Learning, Natural Language Processing, and Image Processing. My research interests include advancing solutions in computer vision, natural language processing, and low-resource language processing. Previously, I worked as a Java developer at Business Automation Limited. Guided by the motto, "Education and research pave the way for innovation," I aim to bridge theory with real-world applications in AI and software development.

Education

2018- 2022 **B.Sc in Computer Science & Engineering (CSE) ,Bangladesh University of Business & Technology (BUBT)** .
Thesis title: *Emotion Detection and Prediction of Human Psychology Using Machine Learning Algorithm* .
Result : 3.44 out of 4 .

Employment History

2022 – 2023 **JAVA developer (young professional).** Business Automation Limited, Mirpur DOHS, Dhaka, Bangladesh. As a Java Developer at Business Automation Limited, I worked on services for the Dutch Bangla Bank and government projects, specializing in API development and backend development for platforms like Gas Monkey. I led Cypress automation testing to ensure software quality and gained experience with Spring Boot frameworks.

2023 – Continue **Python Instructor - Remote, India**. As a remote Python instructor in India, I create comprehensive coding courses for an online learning portal and conduct feedback sessions to assist students with any challenges they encounter. This approach helps ensure effective learning and resolution of doubts.

2022 – Continue **Freelance Researcher**. As a freelance researcher, I've collaborated on various projects with international professors, gaining invaluable experience in research methodologies and analytical skills. These unpaid partnerships have expanded my academic network and provided exposure to diverse scholarly perspectives.

Skills

Languages	English, Bengali.
Coding	Java, PHP, Python, C, C++ .
Problem Solving	LeetCode
Databases	MySQL, Oracle .
Web Dev	HTML, CSS, JavaScript, Apache Web Server, Tomcat Web Server.
Software Testing.	Cypress.
Machine Learning	Supervised Learning (Regression, Classification), Unsupervised Learning (Clustering), Deep Learning (ANN, CNN, RNN, LSTM, Transfer Learning), Ensemble Learning (Bagging, Boosting, Stacking), NLP (Data Preprocessing, Word Embeddings, Transformer), Interpretable Machine Learning (SHAP, LIME, Grad-CAM)

Skills (continued)

Misc. — Microsoft Office, LaTeX, Overleaf, Android Studio, Anaconda, Pycharm, Intellij, VS-code, Google Coolab

Area of Research Interest

- Machine Learning, Deep Learning, Natural language processing (LLM, Generative AI, Text Analysis), Image Processing (Classification, Object Detection, Medical Image Analysis), Cyber-security, Cognitive AI, Explainable artificial intelligence (XAI)

Research Publications

Journal Articles

1. A. Al Maruf, A. J. Abidin, M. M. Haque, *et al.*, "Hate speech detection in the Bengali language: A comprehensive survey," *Journal of Big Data*, vol. 11, no. 1, p. 97, 2024, **Scopus Q1**.
2. A. A. Maruf, S. H. Fahim, R. Bashar, R. A. Rumy, S. I. Chowdhury, and Z. Aung, "Classification of freshwater fish diseases in Bangladesh using a novel ensemble deep learning model: Enhancing accuracy and interpretability," *IEEE Access*, vol. 12, pp. 96 411–96 435, 2024, **Scopus Q1**. DOI: `10.1109/ACCESS.2024.3426041`.
3. A. A. Maruf, F. Khanam, M. M. Haque, Z. M. Jiyad, M. Mridha, and Z. Aung, "Challenges and opportunities of text-based emotion detection: A survey," *IEEE Access*, vol. 12, pp. 18 416–18 450, 2024, **Scopus Q1**.
4. A. A. Maruf, Z. M. Ziyad, M. M. Haque, and F. Khanam, "Emotion detection from text and sentiment analysis of Ukraine Russia war using machine learning technique," *International Journal of Advanced Computer Science and Applications*, vol. 13, no. 12, 2022, **Scopus Q3**. DOI: `10.14569/IJACSA.2022.01312101`.

Conference Proceedings

1. A. J. Abidin, M. K. Siam, A. Al Maruf, R. A. Rumy, and Z. Aung, "Leveraging the transfer learning techniques with data augmentation to improve Bangladeshi crop diseases diagnosis," in *The 7th International Conference on Data Analytics and Cyber Security (DACS)*, **Accepted**, Bihar, India, 2024.
2. A. Al Maruf, A. Golder, A. Al Numan, M. M. Haque, and Z. Aung, "Prediction of heart disease and heart failure using ensemble machine learning models," in *Intelligent Systems*, S. K. Udgata, S. Sethi, and X.-Z. Gao, Eds., Singapore: Springer Nature Singapore, 2024, pp. 481–492, ISBN: 978-981-99-3932-9. DOI: `https://doi.org/10.1007/978-981-99-3932-9_41`.
3. M. M. Haque, A. Al Maruf, A. J. Obeid, H. K. Judi, M. H. Bhuiyan, and Z. Aung, "Exploring elderly activity recognition using deep learning through sensor-enabled android app data collection," in *2024 Third International Conference on Power, Control and Computing Technologies (ICPC2T)*, IEEE, 2024, pp. 582–587.
4. S. Hossain, N. N. I. Prova, M. R. Sadik, and A. Al Maruf, "Enhancing crop management: Ensemble machine learning for real-time crop recommendation system from sensor data," in *2024 International Conference on Smart Systems for Applications in Electrical Sciences (ICSSES)*, IEEE, 2024, pp. 1–6.
5. Z. M. Jiyad, A. Al Maruf, M. M. Haque, M. S. Gupta, A. Ahad, and Z. Aung, "DDoS attack classification leveraging data balancing and hyperparameter tuning approach using ensemble machine learning with XAI," in *2024 Third International Conference on Power, Control and Computing Technologies (ICPC2T)*, IEEE, 2024, pp. 569–575.

6 A. L. Roy, M. K. Siam, N. Noor, S. Jahan, and A. Al-Maruf, "Leveraging gene expression data and explainable machine learning for enhanced early detection of type 2 diabetes," in *The 3rd International Conference on Computing Advancements (ICCA)*, **Accepted**, Dhaka, Bangladesh, 2024.

7 A. L. Roy, M. K. Siam, A. Al Maruf, and Z. Aung, "Efficient language identification of Indian languages using MFCCs and deep learning models," in *The 7th International Conference on Data Analytics and Cyber Security (DACS)*, **Accepted**, Bihar, India, 2024.

8 M. R. Sadik, R. I. Sony, N. N. I. Prova, *et al.*, "Computer vision based Bangla sign language recognition using transfer learning," in *2024 Second International Conference on Data Science and Information System (ICDSIS)*, IEEE, 2024, pp. 1–7.

9 S. Sahajuddin, A.-A. Ahmed, A. Al Maruf, and F. Khanam, "A novel weighted ensemble strategy using transfer learning for enhanced cervical cancer classification," in *The 3rd International Conference on Computing Advancements (ICCA)*, **Accepted**, Dhaka, Bangladesh, 2024.

10 A. Al Maruf, A. Al Numan, M. M. Haque, T. T. Jidney, and Z. Aung, "Ensemble approach to classify spam SMS from Bengali text," in *International Conference on Advances in Computing and Data Sciences*, Springer, 2023, pp. 440–453. DOI: `https://doi.org/10.1007/978-3-031-37940-6_36`.

11 A. A. Maruf, A. Golder, M. S. Naser, A. J. Abidin, A. A. C. Giti, and Z. Aung, "Development of automatic number plate recognition system of Bangladeshi vehicle using object detection and OCR," in *International Conference on Advances in Data-driven Computing and Intelligent Systems*, Springer, 2023, pp. 331–342.

12 A. Al Maruf, M. A.-A. Nayem, M. M. Haque, Z. M. Jiyad, A. M. O. Rashid, and F. Khanam, "A survey on personality prediction," in *Proceedings of the 2nd International Conference on Computing Advancements*, 2022, pp. 407–414. DOI: `https://doi.org/10.1145/3542954.3543012`.

13 A. A. Maruf, M. N. H. Biplob, and F. Khanam, "Covid-19 vaccine sentiment detection and analysis using machine learning technique and NLP," in *International Conference on Machine Intelligence and Emerging Technologies*, Springer, 2022, pp. 401–414. DOI: `https://doi.org/10.1007/978-3-031-34619-4_32`.

Articles Under Review

Journal

- TransembleNet: Enhancing Species and Vector Mosquito Classification through Transfer Learning Base Ensemble Model — 1st author. PLOS ONE. **Scopus Q1** (Under third revision).

- EmoBang: Detecting Emotion From Bengali Texts — 1st author. SN Computer Science. **Scopus Q1** (Under first revision).

- Fostering Trust and Interpretability in Skin Cancer Classification: A Hybrid Framework of Deep Learning and Machine Learning with Explainable AI — 2nd author. Heliyon. **Scopus Q1** (Under first revision).

- Multi-label Emotion and Sentiment Detection from Bengali Text Using Hybrid Deep Learning Model — 2nd author. Computer Speech & Language. **Scopus Q1** (Under review).

- Analysis of Factors Affecting Bangladeshi Students' Academic Performances and Their Predictions Using Machine Learning with Explainable AI Techniques — 1st author. International Journal of Child Care and Education Policy. **Scopus Q1** (Under review).

Research Project

Past

Classification of Freshwater Fish Diseases in Bangladesh Using a Novel Ensemble Deep Learning Model: Enhancing Accuracy and Interpretability. [2]

Problem Statement: The research faces challenges due to limited data samples, hindering model generalization. Issues with accuracy and interpretability also persist, as existing models struggle to provide reliable predictions and transparent decision-making, limiting their practical application.

Methodology: To address data scarcity, we apply data augmentation techniques, including geometric and pixel-level transformations, to expand the dataset and prevent overfitting. An ensemble model is used, where individual models are triggered based on performance-weighted infusion, ensuring that the best-performing models contribute most to the predictions. The weight is calculated by mathematical formulation rather than manually. The ensemble approach improves both accuracy and generalization across varying data conditions. Additionally, Grad-CAM is employed to enhance model interpretability by visualizing the key areas influencing decision-making.

Aim: This research aims to make Bengali aquaculture smarter and more beneficial by developing a system that enables early disease identification, thereby improving sustainability and productivity in the industry.

Past Project

Emotion Detection from Text and Sentiment Analysis of Ukraine Russia War using Machine Learning Technique . [4]

Problem Statement: Emotion detection is often domain-specific, and collecting data on the Ukraine-Russia war is critical. Additionally, emotions can vary depending on context, especially in sensitive topics like this conflict.

Methodology: We employed text preprocessing techniques such as tokenization, lemmatization, and feature extraction methods like Bag-of-Words and TF-IDF. For classification, we used classical machine learning algorithms and an ensemble model with boosting techniques. Sentiment analysis was performed using VADER to assess the public's views on the Ukraine-Russia war.

Aim: The aim of this research is to develop a system for detecting emotion, racism, and sentiment, specifically in the context of the Ukraine-Russia war.

Research Project (continued)

On Going and Under Review

- TransembleNet: Enhancing Species and Vector Mosquito Classification through Transfer Learning Base Ensemble Model — 1st author. PLOS ONE. **Scopus Q1** (Under third revision).
  Problem Statement: Current mosquito vector classification methods face challenges in balancing high accuracy, computational complexity, and handling imbalanced datasets.
  Methodology: This study proposes an efficient ensemble model integrating InceptionV3, ResNet-50, and VGG-16 via transfer learning to optimize both performance and computational cost. By fine-tuning pretrained networks, the model reduces parameter complexity, improving accuracy while managing class imbalance. Evaluation across multiple metrics ensures robustness against the accuracy paradox, making it suitable for real-time applications.
  Aim: This research aims to improve classification accuracy in imbalanced biological and medical datasets through an optimized ensemble model, utilizing transfer learning to balance performance and computational efficiency. The ultimate goal is to enhance predictive reliability for early disease detection and clinical decision-making in resource-constrained environments.